Graphic Era
HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: M.Tech.(CSE) & III
Name of the Paper: Research Methodology & IPR
Paper Code: MCS314
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Discuss the characteristics of a good research problem. Identify and critically evaluate the common errors researchers make when selecting a research problem. CO1

b. Explain the various approaches to investigating solutions for research problems. Provide an example for each approach. CO1

c. How does data collection, analysis, and interpretation influence the outcome of a research problem? Suggest necessary instrumentation techniques for enhancing the reliability of data collection. CO1

Q2. (2X10=20 Marks)

a. Explain the role of plagiarism detection in research. What are the consequences of plagiarism in academic and industrial research? CO2

b. Describe the steps for writing an effective research proposal. How does a review committee assess the quality of a proposal? CO2

c. What are the key components of effective technical writing? Illustrate with examples how clarity and precision can impact the quality of research reports. CO2

Q3. (2X10=20 Marks)

a. Explain the process of patenting under the Patent Cooperation Treaty (PCT). How does international cooperation on Intellectual Property help in innovation? CO3

b. Compare and contrast the concepts of patent, copyright, and design. Provide suitable examples to differentiate them. CO3

c. Discuss the technological and economic significance of intellectual property in fostering innovation and development. CO3

Q4. (2X10=20 Marks)

a. Analyze the scope of patent rights in the context of technology transfer. How do licensing agreements impact innovation? CO4

b. Explain the importance of patent information and databases in research and development. How can they be utilized for strategic decision-making? CO4

c. What are Geographical Indications (GIs)? Discuss their significance in protecting traditional knowledge and promoting regional products. CO4

Q5. (2X10=20 Marks)

a. Discuss the challenges in administering the patent system for emerging technologies like artificial intelligence and biotechnology. CO5

b. Explain the role of IPR in protecting traditional knowledge with suitable case studies. CO5

c. How have universities contributed to the development and protection of intellectual property? Provide specific examples and case study. CO5